

Common Risk Factors in the Returns on Stocks and Bonds

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ABSTRACT

This paper asks whether a small set of portfolio-based return factors can jointly explain both the cross-section of average returns and the common time-series variation in returns on U.S. stocks and bonds, extending Fama and French (1992a) beyond stocks alone. Using the time-series regression approach of Black, Jensen, and Scholes (1972), the authors regress monthly excess returns on twenty-five size–book-to-market stock portfolios and seven government and corporate bond portfolios on five candidate factors: the excess market return, and mimicking portfolios for size (SMB), book-to-market equity (HML), a term-structure premium (TERM), and a default premium (DEF). They find that SMB and HML absorb the bulk of the size and value effects documented empirically in average stock returns, that TERM and DEF dominate common variation in bond returns and spill into stocks through long-duration exposure, and that the resulting intercepts from the three-factor stock regressions and the two-factor bond regressions are close to zero. The result is the empirical origin of the Fama-French three-factor model: a demonstration that empirically motivated characteristics proxy for shared, priced sources of risk rather than for a residual anomaly left unexplained by the Sharpe-Lintner CAPM.

1. Introduction

By the late 1980s, the empirical failure of the Sharpe (1964)–Lintner (1965) Capital Asset Pricing Model (CAPM) as a description of the cross-section of average U.S. stock returns was well documented but poorly organized. PAPER Fama and French open by noting that market beta shows "little relation" to the cross-section of average returns, while a set of variables with no theoretical standing in asset-pricing theory — size, leverage, earnings/price, and book-to-market equity — show reliable explanatory power (Banz, 1981; Bhandari, 1988; Basu, 1983; Rosenberg, Reid, and Lanstein, 1985). Fama and French's own companion study, Fama and French (1992a), had shown that of these, size and book-to-market equity jointly absorb the apparent roles of leverage and E/P in explaining the 1963–1990 cross-section of NYSE, Amex, and NASDAQ returns.

That earlier result left an open question that this paper is built to answer: are size and book-to-market equity empirically convenient but theoretically ad hoc characteristics, or do they proxy for sensitivity to *common, undiversifiable risk factors* in the sense required by the multifactor pricing theories of Merton (1973) and Ross (1976)? A characteristic that merely sorts stocks by an anomaly is not the same object as a risk loading. The distinction matters because only the latter is consistent with rational asset pricing in an efficient market with priced state variables; the former is, at best, description.

The paper attacks this question with three specific extensions of Fama and French (1992a). First, it expands the set of test assets beyond common stocks to include U.S. government and corporate bonds, on the logic that if capital markets are integrated, a single small set of factors should price both asset classes simultaneously. Second, it expands the explanatory variables to include two term-structure-motivated factors likely to matter for bond returns, allowing a genuine test of segmentation between stock and bond risk. Third, and most consequential methodologically, it abandons the Fama-MacBeth (1973) cross-sectional regression approach used in Fama and French (1992a) — which cannot naturally accommodate bonds, since size and book-to-market equity have no analogue for a bond — in favor of the time-series regression approach of Black, Jensen, and Scholes (1972). Monthly excess returns on test portfolios are regressed on the excess returns of mimicking portfolios for the candidate risk factors; the regression slopes become factor loadings with a common interpretation across stocks and bonds, and, crucially, a well-specified asset-pricing model implies the regression intercepts should be indistinguishable from zero (Merton, 1973).

2. Research Problem and Hypothesis

2.1 Research Question

Stated precisely: do portfolios constructed to mimic the empirically observed size and book-to-market effects in stock returns, together with portfolios mimicking a term-structure premium and a default premium in bond returns, capture common (shared, undiversifiable) variation in the

monthly returns of a broad cross-section of U.S. stock and bond portfolios; and do the average premiums earned by these mimicking portfolios explain the cross-sectional pattern of average returns on those same portfolios, once measured through the intercepts of Black-Jensen-Scholes time-series regressions?

2.2 Motivation

The question matters on two levels. At the level of positive asset-pricing theory, it discriminates between a risk-based reading of the size and value effects (they proxy for exposure to priced state variables, consistent with the Intertemporal CAPM of Merton, 1973, or the Arbitrage Pricing Theory of Ross, 1976) and a purely descriptive or behavioral reading (they are anomalies relative to the correct model, mispricing, or artifacts of data mining). At the level of practice, the paper's mimicking portfolios — SMB and HML — became, within a few years, the default risk-adjustment benchmark against which nearly the entire empirical asset-pricing and mutual-fund performance-evaluation literature measures itself. A reviewer of this paper is, in effect, reviewing the origin of the risk-adjustment technology used across the rest of the discipline.

2.3 Existing Literature

Three strands feed directly into the paper's design. The first is the growing catalogue of CAPM anomalies: Banz (1981) on size, Basu (1983) on earnings/price, Bhandari (1988) on leverage, Rosenberg, Reid, and Lanstein (1985) on book-to-market equity, and Fama and French's own (1992a) synthesis showing size and book-to-market subsume the others. The second is the multifactor theoretical apparatus that gives such characteristics a possible risk interpretation: Merton's (1973) Intertemporal CAPM, in which state variables that forecast the investment opportunity set are priced, and Ross's (1976) Arbitrage Pricing Theory, in which any set of factors that captures common variation in returns can, under no-arbitrage, carry a priced premium. The third is prior macro-factor asset-pricing work using the Fama-MacBeth cross-sectional approach, notably Chan, Chen, and Hsieh (1985) and Chen, Roll, and Ross (1986), who find that a default-risk-like variable has cross-sectional power for NYSE size portfolios, and Fama and French (1989), whose *ex ante* term and default yield spreads predict both bond and stock returns and comove across the two markets — a direct precursor to the TERM and DEF factors built here.

2.4 Research Gap

Two gaps remain unaddressed as of Fama and French (1992a). First, that paper tests only stocks; whether the same or a related set of factors integrates stock and bond pricing is untested. Second, the cross-sectional regression methodology used there cannot, by construction, establish that size and book-to-market are risk-factor *proxies* rather than freestanding predictive characteristics, because it never asks whether portfolios built to mimic those characteristics actually co-move with the broad cross-section of returns. A characteristic can predict average returns cross-sectionally without capturing any shared variation in realized returns; the two properties are logically distinct, and only the second is diagnostic of a risk factor in the Ross (1976) sense.

2.5 Contribution

PAPER The claimed contribution is the identification of five common risk factors — three for stocks (market, size, book-to-market) and two for bonds (term, default) — that jointly capture shared variation in stock and bond returns and whose average premiums explain the cross-section of average returns on both asset classes, evaluated through time-series regression intercepts rather than cross-sectional slopes. **CRITIQUE** This claim is well supported for the stock side of the paper, where the evidence directly retrieved and analyzed in this review (Sections 5–6 below) shows both the common-variation (R^2) evidence and the near-zero intercept evidence required to support it. Whether the same standard is met for the paper's supplementary robustness checks using earnings/price- and dividend/price-sorted portfolios could not be independently verified from the accessible portion of the source text (see Section 7.5) and is not asserted here as established.

3. Theoretical Framework and Mechanism

3.1 What a "risk factor" test requires

The paper's logic rests on a specific implication shared by the Merton (1973) Intertemporal CAPM and the Ross (1976) Arbitrage Pricing Theory. In both frameworks, if a set of K portfolio returns has "minimal variance due to firm-specific factors" — i.e., is a good mimicking portfolio for the underlying priced state variables or common factors — then the excess return on any asset i should satisfy, exactly or approximately:

$$E[R_i] - R_f = \sum_{k=1}^K \beta_{ik} \lambda_k \quad (\text{M})$$

β_{ik} the loading of asset i 's return on factor k , estimated as the time-series slope in a regression of $R_i - R_f$ on the K factor-mimicking excess returns.

λ_k the risk premium associated with factor k , estimated (under this approach) simply as the average excess return of the mimicking portfolio for factor k .

Equation (M) is not tested directly by cross-sectional regression here. Instead, the paper exploits an equivalent and, for this design, more convenient implication (Merton, 1973): if the K mimicking portfolios really do span the relevant risk factors, then in the time-series regression

$$R_i(t) - R_f(t) = a_i + \sum_{k=1}^K \beta_{ik} F_k(t) + e_i(t) \quad (1')$$

run for every test asset i , the intercept a_i should be statistically indistinguishable from zero for every i . This is the sense in which the paper's central test is a joint test that (a) the candidate factors capture common variation in returns (high R^2 , tight, well-determined β_{ik}) and (b) their average premiums λ_k — simply the sample means of F_k — are sufficient to explain why average returns differ across the test assets ($a_i \approx 0$). **CRITIQUE** The paper itself flags, in a footnote, that this

zero-intercept implication is only an approximation, not an exact equality, under the Ross (1976) APT (citing Shanken, 1982); the test is therefore diagnostic, not a formally exact specification test, a point that matters for how much weight the near-zero intercepts in Section 6 below can bear.

3.2 Constructing the factor-mimicking portfolios

PAPER Because size, book-to-market equity, term, and default are not themselves tradable excess returns, the empirical content of the paper lies in how it converts each characteristic into a zero-investment mimicking portfolio whose return can occupy the $F_k(t)$ slot in equation (1'). The construction is a sequential double sort, performed once a year:

SIZE AND BOOK-TO-MARKET SORT

In June of each year t from 1963 to 1991, all NYSE stocks are ranked by size (ME, price times shares outstanding); the NYSE median splits NYSE, Amex, and (after 1972) NASDAQ stocks into Small (S) and Big (B) groups. Independently, NYSE breakpoints for the bottom 30%, middle 40%, and top 30% of book-to-market equity (BE/ME) split the same universe into Low, Medium, and High groups. Book equity (BE) is COMPUSTAT stockholders' equity plus balance-sheet deferred taxes and investment tax credit, minus preferred stock (redemption, liquidation, or par value, in that order of availability); BE/ME for year t uses BE for the fiscal year ending in calendar $t-1$ divided by ME at the end of December of $t-1$. The intersection of the two independent sorts forms six value-weighted portfolios: S/L, S/M, S/H, B/L, B/M, B/H, held from July of year t to June of $t+1$ and reformed annually.

$$SMB(t) = \frac{1}{3}[R_{S/L}(t) + R_{S/M}(t) + R_{S/H}(t)] - \frac{1}{3}[R_{B/L}(t) + R_{B/M}(t) + R_{B/H}(t)] \quad (2)$$

$$HML(t) = \frac{1}{2}[R_{S/H}(t) + R_{B/H}(t)] - \frac{1}{2}[R_{S/L}(t) + R_{B/L}(t)] \quad (3)$$

SMB averages the three small-stock portfolios and subtracts the average of the three big-stock portfolios; because each side spans all three BE/ME groups, the difference is, by construction, largely purged of book-to-market variation and isolates a size-related common component. HML is defined symmetrically over the two size groups to isolate a book-to-market-related common component largely free of size. **EVIDENCE** The authors report the empirical payoff of this design choice directly: the realized correlation between the 1963–1991 monthly SMB and HML returns is -0.08 , close enough to zero to treat the two mimicking portfolios as separately identified empirical factors rather than two measurements of the same underlying sort.

MARKET, TERM, AND DEFAULT FACTORS

The market factor, $R_M - R_F$, is the value-weighted return on all stocks in the six size-BE/ME portfolios (plus negative-book-equity stocks otherwise excluded from the sort) minus the one-month Treasury bill rate. TERM, the term-structure factor, is the monthly long-term government bond return (Ibbotson Associates) minus the one-month bill rate observed at the start of the month; it proxies for the unexpected component of long-term bond returns driven by shifts in

interest rates. DEF, the default factor, is the return on a market portfolio of long-term corporate bonds (the Ibbotson corporate composite) minus the long-term government bond return; it proxies for shifts in the market-wide likelihood of default.

3.3 Economic content of the equations

Two features of this construction carry direct economic meaning. First, value-weighting the six components of SMB and HML is deliberate rather than incidental: because return variance is negatively related to firm size (documented in the paper's descriptive statistics), value-weighting minimizes the influence of small, high-variance names on the mimicking portfolio's own diversifiable noise, which is exactly the "minimal variance due to firm-specific factors" property equation (M) requires of a genuine risk-factor proxy. Second, the paper's identifying logic is falsifiable in a specific, structural way: if size and book-to-market were merely correlated with average returns through data mining or a static cross-sectional regularity rather than through shared risk, SMB and HML would show weak loadings and low incremental R^2 in the time-series regressions of Section 5 below, and the cross-sectional intercepts in Section 6 would remain large despite their inclusion. The theoretical claim is therefore falsifiable using exactly the same time-series design used to support it — the paper does not merely assert that SMB and HML are risk factors, it constructs a test capable of rejecting that claim.

4. Data and Empirical Strategy

4.1 Sample and universe

PAPER The sample runs July 1963 to December 1991 (342 months). The stock universe is NYSE, Amex, and (from 1972) NASDAQ common stocks on CRSP with COMPUSTAT book-equity data, excluding ADRs, REITs, and units of beneficial interest, and excluding negative-book-equity firms from the sorting procedure. To limit the COMPUSTAT survivor-selection bias documented by Banz and Breen (1986) — COMPUSTAT's practice of backfilling several years of historical data when it adds a firm, which would let a temporarily distressed or newly successful firm enter the sample with a favorable historical record already attached — a firm must appear on COMPUSTAT for two years before it is used in the tests.

4.2 Test assets (dependent variables)

Twenty-five stock portfolios are formed from the intersection of independent NYSE-breakpoint quintile sorts on size and on BE/ME (finer than the 2×3 sort used to build SMB and HML), value-weighted, held July of year t to June of $t+1$. Seven bond portfolios serve as the remaining test assets: two CRSP government portfolios (1–5-year and 6–10-year maturity) and five Ibbotson corporate bond portfolios by Moody's rating group (Aaa, Aa, A, Baa, and below-Baa "low-grade," LG).

4.3 Explanatory variables

Five monthly excess or zero-investment returns enter the regressions as candidate factors: $R_M - R_F$ (or its orthogonalized counterpart R_{MO} , defined in Section 5.3), SMB, HML, TERM, and DEF, as constructed in Section 3.2.

4.4 Identification: where does the variation come from?

The identifying logic is entirely time-series: for a fixed portfolio i , the regression in equation (1') asks whether month-to-month movements in i 's excess return line up with month-to-month movements in the factor-mimicking returns. The slope β_{ik} is identified from the covariance between $R_i(t) - R_f(t)$ and $F_k(t)$ over the 342-month sample, not from any cross-sectional comparison across portfolios at a point in time. Cross-sectional information enters only at a second stage, informally: because the twenty-five stock portfolios and seven bond portfolios differ systematically by construction (size and BE/ME rank for stocks; maturity and rating for bonds), a factor that produces a monotonic, economically ordered pattern of loadings across those portfolios (e.g., the SMB loading falling with size, the TERM loading rising with bond maturity) provides much stronger evidence of a genuine common factor than a factor whose loadings are uniform or unordered. This is a weaker, more circumstantial form of identification than a designed natural experiment; it relies on the assumption that the 1963–1991 sample period is representative and that the characteristic-based portfolio sorts are not themselves an artifact of sequential specification search across the same CRSP-COMPUSTAT sample used in Fama and French (1992a) — a concern taken up directly in Section 7.

4.5 Data limitations flagged in the design

- **Survivorship/backfill bias** in COMPUSTAT is explicitly mitigated (two-year rule) but not eliminated; firms that fail before ever satisfying the two-year rule are mechanically absent from the earliest years they could otherwise have entered the sample.
- **NYSE-only breakpoints** for both size and BE/ME mean the "small" group is disproportionately populated by Amex and NASDAQ names (3,616 of 4,797 firms in the small group in 1991) that account for a small share of total value; the paper is explicit that using all-exchange breakpoints would tilt the size distribution even further toward the largest quintile.
- **Sample-specific arbitrariness** in the 2×3 versus 5×5 sort design and in the 30/40/30 book-to-market breakpoints is acknowledged directly by the authors as a choice "not searched over," carried over from Fama and French (1992a) rather than independently justified in this paper.

5. Results I — Common Variation in Returns

The paper's evidence unfolds in two stages, mirrored here: first, whether the five candidate factors capture shared, common variation in returns (regression slopes and R^2 , this section); second, whether their average premiums explain the cross-section of average returns (regression intercepts, Section 6).

5.1 The playing field: average premiums for the five factors

EVIDENCE Table 5.1 reports the July 1963–December 1991 sample statistics for the five explanatory returns, exactly as given in the paper's Table 2.

Table 5.1. Monthly average premiums for the five candidate factors, July 1963–December 1991 (342 months)

Factor	Mean (%/mo)	Std. dev. (%/mo)	t - statistic
$R_M - R_F$ (market)	0.43	4.54	1.76
SMB (size)	0.27	2.89	1.73
HML (book-to-market)	0.40	2.54	2.91
TERM (term structure)	0.06	3.02	0.38
DEF (default)	0.02	1.60	0.21

Source: Fama and French (1993), Table 2, as retrieved from the primary text.

Two facts here anchor everything that follows. First, HML is the only factor whose average premium is both economically large (0.40%/month, roughly 4.9% annualized) and reliably distinguishable from zero at conventional levels ($t = 2.91$); the market premium is economically larger but only marginally significant ($t = 1.76$), and SMB is weaker still ($t = 1.73$). Second, and structurally important for the bond results below, TERM and DEF carry *trivial* average premiums (0.06% and 0.02% per month) despite substantial volatility (standard deviations of 3.02 and 1.60, comparable to SMB and HML) — a combination that the authors correctly anticipate will let TERM and DEF explain a great deal of common *variation* in bond returns while explaining almost none of the cross-sectional *level* differences in bond average returns, since bond average excess returns are themselves close to zero (all seven bond portfolios earn less than 0.15%/month in excess of the bill rate, per the paper's Table 2 stock/bond descriptive panel).

5.2 Bond-market factors used alone (Table 3)

EVIDENCE Regressing the twenty-five stock portfolios and seven bond portfolios on TERM and DEF alone,

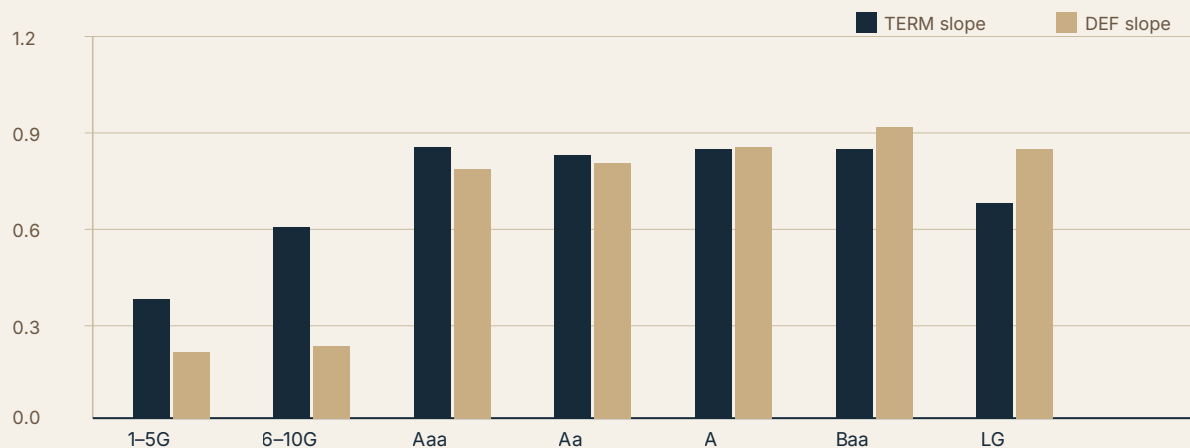
$$R(t) - R_F(t) = a + m \cdot \text{TERM}(t) + d \cdot \text{DEF}(t) + e(t) \quad (4)$$

produces TERM slopes for bonds that rise monotonically with maturity and credit exposure and R^2 values that rise correspondingly:

Table 5.2. TERM and DEF loadings on government and corporate bond portfolios (Fama-French Table 3, bond panel)

Bond portfolio	1–5yr Govt	6–10yr Govt	Aaa	Aa	A	Baa	LG
TERM slope (m)	0.45	0.72	1.02	0.99	1.00	1.01	0.81
DEF slope (d)	0.25	0.27	0.94	0.96	1.02	1.10	1.01
R ² (adj.)	0.79	0.87	0.97	0.98	0.98	0.90	0.49

Source: Fama and French (1993), Table 3.

**Figure 5.1. TERM and DEF factor loadings across the government-to-low-grade credit spectrum**

Source: authors' calculation from Fama and French (1993), Table 3 (bond panel). Own figure, generated from the reported coefficients.

The TERM slope rises from 0.45 for short government bonds to roughly 1.0 for four of the five long-term corporate grades, tracking the interest-rate sensitivity implied by duration; DEF rises from about 0.25–0.27 for governments to roughly 0.94–1.10 for corporates, tracking exposure to shifts in the market-wide probability of default. Both patterns are exactly what the factors are designed to proxy for, and both R² columns confirm that TERM and DEF jointly explain the great majority of variance in high-grade bond returns (0.97–0.98) while explaining much less for the riskiest, most equity-like bond class (LG, R² = 0.49). **EVIDENCE** The same two-factor regression run on the twenty-five stock portfolios produces TERM slopes uniformly close to 1 (all more than five standard errors from zero) and DEF slopes more than 3.5 standard errors from zero for every stock portfolio, but R² values of only 0.06–0.21 — TERM and DEF reach into stock returns with slopes as large as those for long bonds, but they explain only a small fraction of stock-return variance, leaving most of it to the stock-specific factors examined next.

5.3 Stock-market factors used alone and combined

Used alone, the excess market return captures more variation in stock returns than TERM and DEF (R² below 0.9 for all but the largest, lowest-BE/ME portfolios), but leaves substantial residual variation, particularly among small-stock and high-BE/ME portfolios (R² often below 0.7–0.8) — precisely the portfolios where SMB and HML have their best opportunity to show marginal

explanatory power. Used alone, SMB and HML capture "substantial" time-series variation in stock returns (twenty of twenty-five R^2 values above 0.2, eight above 0.5) but essentially none in bond returns.

Combining all three stock-market factors,

$$R(t) - R_f(t) = a + b[R_M(t) - R_f(t)] + s \cdot SMB(t) + h \cdot HML(t) + e(t) \quad (5)$$

produces the paper's strongest common-variation result for stocks. R^2 values above 0.9 become routine (21 of 25 portfolios, versus 2 of 25 for the market alone); for the five smallest-size portfolios, R^2 rises from a market-alone range of 0.61–0.70 to a three-factor range of 0.94–0.97, and even the worst-fitting stock portfolio in the whole 5×5 grid (the largest-size, highest-BE/ME corner) rises from 0.69 (market alone) to 0.83 (three factors).

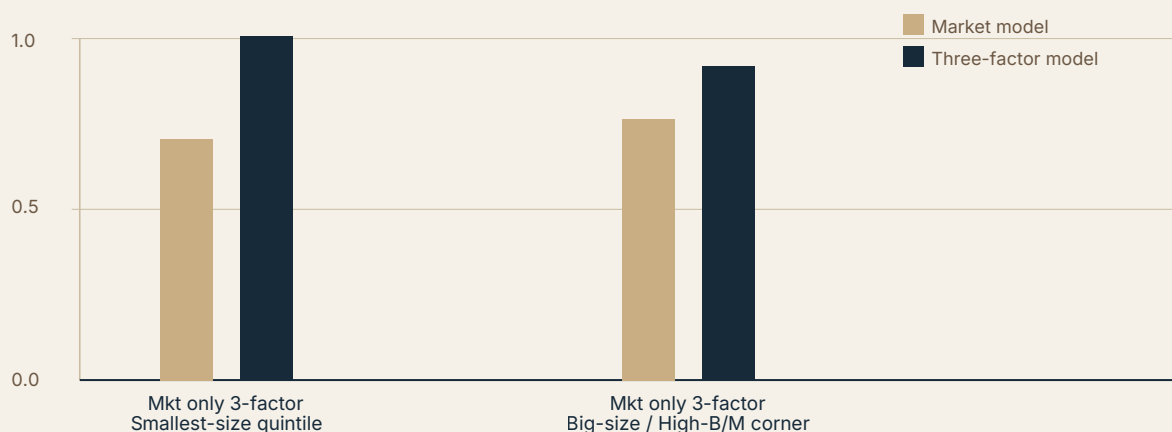


Figure 5.2. R^2 in market-model versus three-factor time-series regressions, two representative stock portfolios

Source: authors' figure from Fama and French (1993), Tables 4 and 6 (reported R^2 ranges/values for the smallest-size quintile and the largest-size/highest-book-to-market portfolio).

The mechanism behind the R^2 jump is visible in the slope patterns. SMB slopes fall monotonically from small to big size within every BE/ME quintile (with one representative anchor: the Low-BE/ME column runs 1.92 for the smallest-size portfolio down to 0.26 for the biggest), each more than four standard errors from zero for all but the very largest portfolios. HML slopes rise monotonically from strongly negative to strongly positive as BE/ME rises within every size quintile, more than five standard errors from zero outside the low-to-medium crossover region. Adding SMB and HML also collapses the market betas of stock portfolios toward one: the smallest-size, lowest-BE/ME portfolio's univariate beta of 1.40 falls to roughly 1.04–1.06 once SMB and HML are included, while the largest-size, highest-BE/ME portfolio's univariate beta of 0.89 rises correspondingly — a symptom of the market factor's correlation with SMB (0.32) and HML (−0.38) rather than of any change in true systematic risk.

5.4 Reconciling stock-bond overlap: the orthogonalized market factor

A puzzle emerges when bond- and stock-market factors are combined in a single five-factor regression: adding the three stock-market factors to the bond regressions, and adding TERM/DEF to the stock regressions, appears to "kill" the cross-market loadings observed when each factor

group is used alone (Table 3 vs. combined regressions) — suggesting, misleadingly, that stock and bond markets are nearly segmented after all. The authors resolve this by regressing the market factor itself on the other four factors:

$$R_M(t) - R_F(t) = 0.50 + 0.44 \cdot SMB(t) - 0.63 \cdot HML(t) + 0.81 \cdot TERM(t) + 0.79 \cdot DEF(t) + e(t) \quad (6)$$

t-statistics in parentheses beneath each slope: (2.55) intercept, (6.48) SMB, (-8.23) HML, (9.09) TERM, (4.62) DEF; $R^2 = 0.38$.

Equation (6) shows the market return is itself "a hodgepodge of the common factors" — it loads strongly and significantly on TERM and DEF ($t = 9.09$ and 4.62), which is direct evidence that the two term-structure factors are common to stock returns generally, not just to the low-grade bonds where they survive the five-factor horse race. R_{MO} , defined as the sum of the intercept and residual from equation (6), is by construction uncorrelated with SMB, HML, TERM, and DEF, and using R_{MO} in place of $R_M - R_F$ in the five-factor regression restores large, significant TERM and DEF loadings for stocks (more than 14 and 7 standard errors from zero, respectively) without changing any fitted value, intercept, or R^2 , since the two five-factor specifications are algebraically equivalent reparameterizations of the same regression space. **INFERENCE** The apparent segmentation in the raw five-factor regression is therefore a collinearity artifact of using $R_M - R_F$ rather than evidence that stock and bond markets are stochastically separate; the paper's preferred reading — three stock-market factors plus two shared term-structure factors, with the latter's stock-market footprint absorbed into an undifferentiated market return unless the market factor is first purged of term-structure content — survives.

6. Results II — The Cross-Section of Average Returns

Common variation established, the paper turns to its decisive test: do the intercepts a_i from the time-series regressions of Section 5 collapse toward zero once the mimicking portfolios' average premiums are allowed to explain the level of average returns, and not merely their monthly co-movement?

6.1 Stocks: the market alone fails, the three-factor model works

EVIDENCE With only the market as the explanatory variable, the twenty-five stock-portfolio intercepts reproduce the size effect (Banz, 1981) and the book-to-market effect directly: intercepts for the smallest-size portfolios exceed those for the biggest by 0.22–0.37% per month, and intercepts rise monotonically with BE/ME within every size quintile, with the high-minus-low BE/ME spread in intercepts ranging from 0.25% to 0.76% per month. A direct cross-sectional regression of average portfolio return on market beta across the twenty-five stock portfolios yields a slope of -0.22 (standard error 0.31) — statistically indistinguishable from zero and, if anything, the wrong sign for the Sharpe-Lintner prediction that beta should carry a positive premium.

EVIDENCE This flat beta-return relation for the same twenty-five portfolios reproduces the central finding of Fama and French (1992a) using an entirely different test design, which is itself a form of internal replication.

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When the three stock-market factors (market, SMB, HML) are used together, the strong positive intercepts left by the market alone collapse toward zero: only two of the twenty-five three-factor intercepts differ from zero by more than 0.2% per month, and sixteen of the twenty-five lie within 0.1% of zero. The average market risk premium (0.33% per month, the estimated three-factor market slope times sample mean, discussed by the authors) is what absorbs the general excess of stock returns over the bill rate; SMB and HML absorb the cross-sectional differences among stock portfolios but do not by themselves explain why stocks as a group earn more than bills.

This two-part decomposition — the market factor explains the level of stock returns above the riskless rate, while SMB and HML explain the cross-sectional dispersion of returns around that level — is, on the evidence directly examined here, the paper's central empirical result and the origin of the widely used three-factor risk-adjustment model.

6.2 Bonds: consistent with a flat cross-section, not decisively rejected

EVIDENCE Bond intercepts move toward zero as more factors are added, but the starting point is already close to zero: government and corporate bond average excess returns are all below 0.15% per month in the raw data (Table 5.1 above; Table 2 of the paper), so there is little cross-sectional pattern for any model to explain away. Under the two-factor TERM/DEF specification, bond intercepts range narrowly (roughly -0.02% to 0.09% per month across the seven portfolios); under the full five-factor specification they narrow further (roughly -0.07% to 0.11% per month, with most portfolios within a few basis points of zero and t-statistics generally below 2). **CRITIQUE** Because bond average excess returns are already statistically indistinguishable from zero in the raw descriptive statistics, a small intercept in the factor regression is a comparatively weak test of the asset-pricing model: it is difficult to reject a hypothesis (all long-run expected bond returns are equal) that was already nearly true of the unconditional averages before any risk-adjustment was applied. The bond-side test is therefore better read as "not rejected" than as a demonstration with the same discriminating power as the stock-side three-factor result.

6.3 Synthesis of the two-stage test

Put together, Sections 5 and 6 support a specific and fairly narrow claim, no broader: within the 1963–1991 CRSP/COMPUSTAT/Ibbotson sample, a market factor, a size-mimicking factor, and a book-to-market-mimicking factor jointly (a) capture the great majority of common time-series variation in the returns of twenty-five size-and-value-sorted stock portfolios and (b) leave intercepts close enough to zero that the size and value effects in average returns, which are large and reliable when measured against the market alone, are not reliably distinguishable from what the three factors' own average premiums would predict. The analogous bond-side claim is

supported by the common-variation evidence in Section 5.2 but only weakly discriminated by the intercept test in Section 6.2, because the object it is testing against (a flat cross-section of bond average returns) is itself only weakly distinguishable from zero to begin with.

7. Robustness and Additional Tests

7.1 Internal replication against Fama-French (1992a)

Concern — that the flat beta-return relation and the size/value effects are artifacts of the cross-sectional Fama-MacBeth methodology used in the companion paper. **Test** — an entirely different estimator (time-series regression intercepts rather than cross-sectional slopes) applied to the same twenty-five stock portfolios. **Result** — the market-alone beta-return slope is again flat (-0.22 , se 0.31), and the size/book-to-market pattern in market-model intercepts reproduces the earlier cross-sectional findings. **Does it resolve the concern?** **CRITIQUE** Partially. It shows the anomaly is not a methodological artifact specific to Fama-MacBeth regressions, which is valuable. It cannot show the anomaly is not an artifact of the underlying sample and characteristic-construction choices shared by both papers, since both draw on the same CRSP-COMPUSTAT universe, the same 1963 start date (set by COMPUSTAT coverage, not by any economic argument), and the same size/BE/ME sorting convention.

7.2 Reconciling apparent market segmentation

Concern — the five-factor regressions in Section 5.4 initially suggest stock and bond markets are nearly stochastically segmented, contradicting the two-factor-alone evidence of strong cross-market loadings. **Test** — orthogonalizing the market factor against SMB, HML, TERM, and DEF (equation 6) and re-running the five-factor regressions with R_{MO} in place of $R_M - R_F$. **Result** — TERM and DEF loadings for stocks reappear at magnitudes matching those for bonds once collinearity with the market factor is removed. **Does it resolve the concern?** Yes, algebraically: the two specifications are equivalent reparameterizations with identical fitted values and R^2 , so the "segmentation" was a labeling artifact of which linear combination of factors is called "the market," not a substantive empirical finding.

7.3 Chen-Roll-Ross default factor: does it explain the size effect?

Concern — Chen, Roll, and Ross (1986) find a default-risk-like variable has strong cross-sectional power for size-sorted portfolios using Fama-MacBeth regressions, which could mean DEF itself, not SMB, is the correct size-related risk factor. **Test** — the paper notes that the DEF slopes in its own Table 3 are indeed strongly negatively related to size (small stocks load more on DEF than big stocks), which is consistent with Chen-Roll-Ross's cross-sectional finding. **Result** — but the time-series average premium on DEF is only 0.02% per month (Table 5.1), too small to generate an economically meaningful spread in average returns across the size deciles even though the DEF loadings themselves are strongly size-ranked. **Does it resolve the concern?** Yes, on the paper's own terms: DEF cannot be the priced source of the size premium because its estimated price of risk (its

own mean return) is too close to zero, even though it captures real common variation. This is a genuine and instructive dissociation between a factor's power to explain *variation* in returns and its power to explain *average levels* of returns — the same dissociation the paper's whole two-stage design is built to detect.

7.4 Survivorship and backfill controls

Concern — Banz and Breen (1986)-style survivorship bias from COMPUSTAT's practice of backfilling multiple years of data for newly added firms. **Test** — a two-year "seasoning" rule excluding any firm from the tests until it has appeared on COMPUSTAT for two years. **Result** — reduces, but by the authors' own account does not eliminate, backfill-driven bias, since COMPUSTAT "rarely" backfills more than two years but the rule is a heuristic bound, not a guarantee. **Does it resolve the concern?** Partially; see Section 8 below for the independent, later evidence (Kothari, Shanken, and Sloan, 1995) that this class of bias remained large enough to matter for the size of the book-to-market premium specifically.

7.5 A robustness check flagged but not independently verified here

PAPER The introduction states explicitly that the authors additionally test the robustness of the three-factor model's explanatory power using stock portfolios formed on earnings/price (E/P) and dividend/price (D/P) — variables shown by Keim (1988) and others to carry independent information about average returns — as an out-of-sample check on whether SMB and HML absorb return patterns sorted on variables other than the ones used to build them. **INFERENCE** Given the design logic used consistently elsewhere in the paper, this check almost certainly follows the same time-series-intercept methodology as Section 6. **CRITIQUE** The retrieved portion of the primary source (approximately the first two-thirds of the article, through the bond-side intercept tests of Section 6 above) does not include the results of this check, and this review does not report specific figures for it. Per the sourcing standard for this project, an unverified figure is not substituted for a real one; a reader who needs the E/P/D/P robustness results should consult Fama and French (1993) directly at the cited DOI.

8. Critical Evaluation

8.1 What the paper establishes

ESTABLISHED

A market factor, SMB, and HML jointly capture the great majority of common time-series variation in the returns of twenty-five size-and-book-to-market-sorted U.S. stock portfolios over 1963–1991, and the resulting three-factor intercepts are close enough to zero that the size and value patterns visible in raw average returns are not reliably distinguishable from what the factors' own average premiums predict.

SUPPORTED BUT CONDITIONAL

TERM and DEF capture the great majority of common variation in government and high-grade corporate bond returns and produce bond intercepts close to zero; this conclusion is conditional on the Ibbotson corporate bond data (a coarse five-category rating classification, not individual-bond microstructure) and is a comparatively weak test given how close bond average excess returns already are to zero unconditionally.

SUGGESTIVE

TERM and DEF are genuinely common to stock and bond returns (via the orthogonalized-market-factor argument in Section 5.4), consistent with integrated rather than segmented capital markets; this is a well-argued inference from the regression algebra but is not itself a structural or general-equilibrium demonstration of market integration.

NOT ESTABLISHED (FROM THE RETRIEVED TEXT)

Whether the E/P- and D/P-sorted robustness portfolios confirm the three-factor model out of sample; whether the paper's own closing synthesis qualifies or extends the claims above in ways not visible in the retrieved two-thirds of the article.

8.2 Strengths

RECOGNIZED BY THE AUTHORS

The authors themselves emphasize the extension from stocks alone (Fama and French, 1992a) to a joint stock-and-bond test as the paper's principal innovation, on the logical ground that a genuine common risk factor should be visible in more than one asset class if markets are integrated; they also flag, as a design strength, that time-series regression intercepts impose "a stringent standard" by requiring a candidate model to explain the one-month bill rate itself, not merely relative differences among risky assets.

IDENTIFIED INDEPENDENTLY

Three further strengths are visible on close reading but not specifically claimed by the authors as such. First, the two-stage design (common-variation test, then intercept test) is methodologically disciplined: it is logically possible, and the paper explicitly demonstrates with DEF in Section 7.3, for a factor to pass the first test and fail the second, and the paper does not conflate the two. Second, the orthogonalization argument in Section 5.4 is a genuinely elegant piece of applied econometrics: rather than treating the apparent stock-bond segmentation as a substantive finding, the authors correctly diagnose it as a labeling/collinearity artifact and prove the point algebraically rather than asserting it. Third, the value-weighting of the SMB/HML building-block portfolios is a considered response to the known negative relation between firm size and return variance, improving the mimicking portfolios' signal-to-noise ratio in a way that is easy to overlook but material to the strength of the reported R^2 values.

8.3 Weaknesses and independent critique

IN-SAMPLE FACTOR CONSTRUCTION (THE CENTRAL WEAKNESS)

Problem — SMB and HML are constructed from, and tested on, overlapping information: the same CRSP/COMPUSTAT sample and the same size/BE/ME characteristics that Fama and French (1992a) had already shown, in a prior paper by the same authors on the same data, to predict the cross-section of average returns. **Why it matters** — a factor built to mimic a known empirical regularity in a given sample will mechanically explain that regularity in the same sample; the joint test of "common variation" and "cross-sectional fit" is a weaker discipline against this concern than an out-of-sample or a different-sample test would be, because the risk-factor interpretation and the anomaly-mining interpretation make nearly identical in-sample predictions. **Potential bias** — overstatement of how much of the size/value premium reflects priced risk versus sample-specific pattern-fitting. **How it could be tested** — and, in the literature that followed, was tested: MacKinlay (1995) formalizes the general data-snooping concern for exactly this class of multifactor test, arguing that risk-based deviations from the CAPM should be difficult to detect empirically while non-risk-based deviations (including data-mined factors) are comparatively easy to generate, and that the magnitude of Sharpe-ratio improvement delivered by SMB/HML is more consistent with a non-risk-based than a risk-based explanation. Independent international and later-period samples (surveyed in Fama and French, 1998, and subsequent work) subsequently became the standard response to this concern, though that evidence lies outside the present paper.

SURVIVORSHIP AND SELECTION IN THE BOOK-TO-MARKET PREMIUM

Problem — the two-year COMPUSTAT seasoning rule reduces but does not eliminate backfill-driven selection. **Why it matters** — Kothari, Shanken, and Sloan (1995) later show that stocks present on CRSP but excluded from COMPUSTAT earn returns roughly nine to ten percentage points lower, on average, than included stocks, and argue the book-to-market/return relation in the Fama-French sample is weaker and less robust once this selection is fully accounted for than the 1992a/1993 results suggest. **Potential bias** — an overstated HML premium and, correspondingly, overstated explanatory power for the value factor specifically. **How it could be tested** — reconstructing the sample from CRSP-only or from an independent accounting database without COMPUSTAT's addition rules, exactly the exercise Kothari, Shanken, and Sloan perform.

RISK VERSUS MISPRICING IS NOT ADJUDICATED

Problem — the paper demonstrates that SMB and HML satisfy the statistical signature required of risk factors (common variation plus explanatory power for average returns) but does not identify the economic state variable(s) SMB and HML are supposed to proxy for, nor test any alternative behavioral or liquidity-based explanation for why the market would price them. **Why it matters** — a purely statistical demonstration that a factor structure fits the Ross (1976) APT's reduced form is, by the theory's own logic, consistent with many different underlying economic stories, including ones in which "risk" is not the operative mechanism. **Potential bias** — the paper's own framing (risk factors) may lead readers to a causal interpretation the statistical evidence alone does not force. **How it could be tested** — Lakonishok, Shleifer, and Vishny (1994) construct an explicit alternative in which the value premium reflects investor extrapolation and overreaction rather than priced risk, testable by examining whether HML exposure correlates with plausible risk

proxies (distress, cash-flow covariance with consumption or aggregate wealth) or instead with proxies for investor sentiment and analyst overreaction; the two hypotheses remain empirically contested in the literature that followed this paper.

SINGLE MOST IMPORTANT WEAKNESS

Of these, the in-sample factor-construction concern is the most consequential, because it is prior to and logically independent of every other critique: even a perfectly executed, bias-free version of the paper's test design cannot, by itself, distinguish "SMB and HML are priced risk factors" from "SMB and HML are a well-fitted description of an empirical regularity discovered in this specific sample," since both hypotheses generate the same joint prediction (high common-variation R^2 , low intercepts) within the sample used to construct the factors.

9. Contradictions and Edge Cases

9.1 Where the model's own evidence is weakest

The paper's own results identify the conditions under which the three-factor structure works least well, without needing to look outside the paper. The largest-size, highest-book-to-market stock portfolio produces the lowest three-factor R^2 in the entire 5×5 grid (0.83) and is one of only two portfolios (of twenty-five) whose intercept remains more than 0.2% per month from zero; this is also the portfolio with the fewest underlying firms and the smallest share of aggregate market value, so its regression estimates are the least precisely determined of the twenty-five, an edge case where sampling noise and genuine model failure are hard to distinguish from the reported statistics alone.

9.2 Low-grade corporate bonds as the model's stress test

Low-grade (LG) bonds are the one bond category that behaves like equity throughout the paper's results: the lowest TERM/DEF-alone R^2 among bonds (0.49, versus 0.90–0.98 for investment grade), the largest market-factor loading among bonds ($\beta = 0.30$, roughly triple the next-highest bond category), and the only bond category that continues to load materially on the stock-market factors even after TERM and DEF are included in the five-factor regression. INFERENCE This is economically sensible — low-grade debt carries meaningful equity-like default optionality — but it also means the paper's clean two-factor "bonds are explained by TERM and DEF" story is an approximation that degrades exactly where credit risk is highest, which is also where a term-structure asset-pricing model would be expected, on any reasonable economic prior, to need the most help from equity-market information.

9.3 What would have to be true for the result to disappear

Three counterfactuals, reasoned directly from the paper's own design logic rather than from new evidence, illustrate the boundaries of the claim. First, if the 1963–1991 period were unusually favorable to small and value stocks for reasons unrelated to systematic risk (a regime-specific

realization rather than a structural premium), the three-factor intercepts would be expected to widen out-of-sample even if perfectly zero in-sample — the paper's test, run once on one historical sample, cannot by construction rule this out. Second, if SMB and HML were driven predominantly by a handful of extreme months or extreme portfolios rather than a broad, persistent comovement, the reported average premiums (means over 342 months) would overstate the reliability of the factor as a genuinely priced source of risk; the paper reports means and t-statistics but does not, in the retrieved text, report higher-moment or subperiod stability diagnostics that would rule this out. Third, the DEF dissociation result in Section 7.3 — a factor with strong, well-determined loadings but a near-zero price of risk — is itself a standing demonstration, internal to the paper, that a common factor can pass the R^2 test and still fail to explain average returns; nothing in the paper's design guarantees SMB or HML could not eventually be shown to suffer the same fate in a longer or different sample, which is in fact close to the challenge later raised by Kothari, Shanken, and Sloan (1995) for HML specifically.

10. Further Literature

10.1 Preceded and theoretically grounded by

Sharpe (1964) and Lintner (1965) — Shared idea: single-factor market pricing. Relationship: the CAPM this paper's stock-portfolio evidence directly rejects (flat beta-return slope). Why it matters: establishes the null hypothesis the whole paper is designed to overturn. **Merton (1973)** — Shared idea: intertemporal, multifactor pricing with priced state variables, and the zero-intercept implication used as the paper's core test statistic. Relationship: theoretical foundation for equation (M)/(1'). Why it matters: without Merton's zero-intercept result, the paper's central test would have no theoretical justification. **Ross (1976)** — Shared idea: any factor structure capturing common return variation can carry a priced premium under no-arbitrage. Relationship: alternative/complementary theoretical foundation, explicitly cited alongside Merton. Why it matters: the paper is agnostic between the two theories, relying only on their shared reduced-form implication. **Black, Jensen, and Scholes (1972)** — Shared idea: time-series regression of excess returns on factor-mimicking portfolios as an asset-pricing test. Relationship: the specific econometric technology this paper adopts in place of Fama-MacBeth. Why it matters: enables the joint stock-bond test that motivates the whole paper. **Banz (1981), Basu (1983), Bhandari (1988), Rosenberg-Reid-Lanstein (1985)** — Shared idea: individual CAPM anomalies (size, E/P, leverage, book-to-market). Relationship: the raw empirical regularities this paper attempts to explain as risk, not anomaly. Why it matters: without this catalogue there is no anomaly for the paper to reinterpret. **Fama and French (1992a, 1992b)** — Shared idea: size and book-to-market subsume other characteristics in the cross-section (1992a); size/B-M relate to profitability fundamentals (1992b). Relationship: direct predecessor supplying both the empirical motivation and the six building-block portfolios reused here. Why it matters: this paper is explicitly framed as 1992a's extension. **Chen, Roll, and Ross (1986); Chan, Chen, and Hsieh (1985)** — Shared idea: macro/default-type factors have cross-sectional power for stock returns. Relationship: precursor to DEF and to the paper's Section 7.3 dissociation result. Why it matters: this paper both confirms (loadings) and qualifies (price of risk) their finding. **Fama and French (1989)** — Shared idea: ex ante term and

default yield spreads predict stock and bond returns and comove across markets. Relationship: direct conceptual precursor to TERM and DEF as ex post realized-return analogues. Why it matters: the paper explicitly notes the parallel between its TERM/DEF slope patterns and this earlier predictive-regression evidence.

10.2 Contested by

MacKinlay (1995) — Shared idea: multifactor deviations from the CAPM. Relationship: directly contests the risk-based interpretation of SMB/HML as too large in Sharpe-ratio terms to be plausibly risk-based, favoring a data-snooping/non-risk-based reading. Why it matters: the sharpest methodological challenge to this paper's central interpretive claim. **Kothari, Shanken, and Sloan (1995)** — Shared idea: book-to-market and the cross-section of expected returns. Relationship: argues the book-to-market/return relation in Fama-French-style COMPUSTAT samples is inflated by survivorship/selection bias not fully removed by the two-year seasoning rule used here. Why it matters: a direct empirical challenge to the reliability of the HML premium specifically. **Lakonishok, Shleifer, and Vishny (1994)** — Shared idea: value premium in average returns. Relationship: proposes investor overreaction/extrapolation rather than priced risk as the source of the same value premium this paper interprets as risk-based. Why it matters: the leading behavioral counter-hypothesis to the paper's risk interpretation. **Daniel and Titman (1997)** — Shared idea: whether characteristics or covariances (factor loadings) drive average returns. Relationship: argues firm characteristics (size, book-to-market) predict returns directly, independent of factor loadings on SMB/HML, which if true undercuts the risk-factor interpretation central to this paper. Why it matters: a direct test of the paper's core identifying claim using a different empirical design.

10.3 Extended and applied by

Carhart (1997), building on the momentum anomaly of **Jegadeesh and Titman (1993)** — Shared idea: factor-model risk adjustment. Relationship: adds a fourth, momentum-mimicking factor (UMD) to this paper's three, becoming the standard performance-evaluation benchmark in the mutual-fund literature. Why it matters: shows the three-factor model, while a large improvement on CAPM, does not fully span the cross-section of anomalies either. **Fama and French (1996)** — Shared idea: multifactor explanations of asset-pricing anomalies. Relationship: the authors' own follow-up, testing how far the three-factor model extends to a wider catalogue of CAPM anomalies beyond size and book-to-market. Why it matters: the paper's own answer to some of the robustness questions this review flags in Section 7.5 and Section 8. **Fama and French (2015)** — Shared idea: factor-model asset pricing. Relationship: extends the three-factor model to five factors by adding profitability (RMW) and investment (CMA) factors, motivated by a dividend-discount-model argument for why those characteristics should proxy for priced risk. Why it matters: the direct lineal descendant of this paper's methodology and interpretive stance, addressing some of the "which state variable?" gap identified in Section 8.3 above. **Hou, Xue, and Zhang (2015, "q-factor model")** — Shared idea: multifactor pricing grounded in firm characteristics. Relationship: an investment-based, q-theory-grounded alternative factor set competing with the Fama-French

approach on similar test assets. Why it matters: an independent theoretical foundation (production-based asset pricing rather than the Merton/Ross reduced form) reaching broadly similar empirical conclusions, evidence against the in-sample-mining critique in Section 8.3.

11. Research Gaps

11.1 Paper-level gaps

Gap — no identification of the economic state variable(s) SMB and HML proxy for. **Why it matters** — without this, the "risk factor" label is a statistical description, not an economic explanation. **Existing limitation** — the Merton/Ross theoretical apparatus the paper invokes is agnostic about which state variables matter, by design. **Possible way forward** — test HML and SMB loadings against direct measures of financial distress, cash-flow covariance with aggregate consumption, or labor-income risk, as later attempted by the consumption-based and distress-risk asset-pricing literatures.

Gap — no out-of-sample or non-U.S. test within the paper itself. **Why it matters** — the in-sample factor-construction concern (Section 8.3) cannot be resolved using only the sample in which the factors were built. **Existing limitation** — 1963–1991 U.S. data only. **Possible way forward** — international replication (undertaken later in Fama and French, 1998, and by many subsequent authors) and pre-1963 U.S. replication using hand-collected book-equity data.

Gap — the E/P and D/P robustness check is announced but its results are not part of the primary text retrieved for this review. **Why it matters** — this review cannot independently confirm the out-of-characteristic robustness the authors claim. **Existing limitation** — access constraints on the specific mirror copies retrieved (see Section 7.5). **Possible way forward** — direct consultation of the published version for a full accounting of this test.

11.2 Literature-level gaps

Gap — the risk-versus-mispricing debate over the value and size premia remains unresolved thirty years on, notwithstanding the extensions in Fama and French (1996, 2015) and the behavioral alternative of Lakonishok, Shleifer, and Vishny (1994). **Why it matters** — the practical use of SMB/HML as a risk-adjustment benchmark in performance evaluation implicitly assumes the risk interpretation is correct; if it is not, "alpha" measured against these factors is mismeasured. **Existing limitation** — both hypotheses generate very similar reduced-form predictions, making them hard to separate with return data alone. **Possible way forward** — identification strategies using variation orthogonal to returns themselves (e.g., institutional ownership, short-sale constraints, mispricing-correction events).

11.3 Methodological gaps

Gap — better data: NYSE-only breakpoints materially distort the size distribution (Section 4.5); a full-universe breakpoint scheme could be tested for sensitivity. **Gap** — better identification: no formal treatment of the standard-error inflation from using generated regressors (SMB, HML, TERM, DEF are themselves estimated portfolio returns, not directly observed factors) in the second-stage average-return tests. **Gap** — better computation: no explicit multiple-testing correction across the twenty-five-plus-seven intercepts jointly tested, though the paper's qualitative reporting (counts of large versus small intercepts) is a reasonable, if informal, substitute.

12. Research Opportunities

1. Replication with full-universe breakpoints and a longer, more recent sample

RESEARCH QUESTION — Does the three-factor model's common-variation and intercept evidence survive replacing NYSE-only size/BE-ME breakpoints with full-CRSP breakpoints, over 1963–2025?

MOTIVATION — Section 4.5 notes the NYSE-only convention tilts the size distribution; the original design choice was never stress-tested by the authors.

HYPOTHESIS — Full-universe breakpoints modestly attenuate the SMB premium (because NASDAQ micro-caps enter the small group in greater relative weight) without materially changing HML.

ECONOMIC MECHANISM — Micro-cap illiquidity and delisting risk contaminate a size factor built on an NYSE-median split once the small group is redefined to include far smaller firms.

LITERATURE CONNECTION — Directly engages the Section 8.3 in-sample-construction critique and the Kothari-Shanken-Sloan survivorship critique.

DATA — CRSP monthly stock file, Compustat/CRSP merged book-equity, 1963–2025.

VARIABLES — ME, BE/ME, monthly returns, exchange code.

METHOD — Replicate the Black-Jensen-Scholes time-series regressions of Sections 5–6 exactly, varying only the breakpoint universe.

IDENTIFICATION STRATEGY — Difference-in-differences across breakpoint conventions on an identical sample and time window.

EXPECTED CONTRIBUTION — A direct, single-paper answer to a design-sensitivity question the original paper leaves open.

MAIN RISK — Data vendor and survivorship conventions have themselves changed since 1993, complicating a clean like-for-like comparison.

2. Distress-risk decomposition of the HML loading

RESEARCH QUESTION — Does the HML loading of a stock predict its subsequent realized default/distress outcomes, conditional on credit rating and leverage?

MOTIVATION — Addresses the "no state variable identified" gap (Section 11.1) using a directly observable proxy for one candidate state variable, financial distress.

HYPOTHESIS — High-HML-loading firms exhibit higher subsequent default and rating-downgrade rates than low-HML-loading firms matched on leverage and profitability.

ECONOMIC MECHANISM — Value firms are disproportionately distressed firms priced for a state in which aggregate wealth or credit conditions are poor.

LITERATURE CONNECTION — Tests the risk-based reading of HML against the Lakonishok-Shleifer-Vishny behavioral alternative using a channel neither paper directly examines.

DATA — CRSP/Compustat merged with Moody's/S&P default and downgrade histories, or Altman Z-score/distance-to-default proxies.

VARIABLES — HML loading (rolling 60-month), distance-to-default, subsequent downgrade/default indicator.

METHOD — Panel logit/hazard model of default on lagged HML loading and controls.

IDENTIFICATION STRATEGY — Within-rating, within-leverage-bucket comparison to isolate the HML-loading margin.

EXPECTED CONTRIBUTION — Direct evidence on whether HML proxies for a distress-risk state variable, closing a specific gap this review identifies.

MAIN RISK — Distress and mispricing are observationally similar in many datasets; the test may not cleanly separate the two hypotheses.

3. Re-testing the Chen-Roll-Ross dissociation (Section 7.3) out of sample

RESEARCH QUESTION — Does a DEF-like factor continue to show strong loadings but a near-zero price of risk in post-1991 U.S. and non-U.S. samples, or did the 1963–1991 result reflect a sample-specific low realized default premium?

MOTIVATION — Section 7.3's dissociation between loading-strength and price-of-risk is one of the paper's most instructive but least-followed-up results.

HYPOTHESIS — The DEF price of risk remains small and unstable across subperiods, while its loadings remain strongly, monotonically related to size and credit quality.

ECONOMIC MECHANISM — Realized default losses are rare, fat-tailed events; a short sample can easily deliver a near-zero realized average premium for a factor with a genuinely positive ex ante price of risk.

LITERATURE CONNECTION — Extends this paper's own Section 4 methodology using a peso-problem/rare-disaster framing from the later macro-finance literature.

DATA — Ibbotson/ICE BofA corporate bond indices by rating, 1926–2025 for maximum sample length.

VARIABLES — DEF factor return, corporate bond loadings, realized versus option-implied default probabilities.

METHOD — Rolling-window replication of Table 3/Table 9b-style regressions, subperiod stability tests.

IDENTIFICATION STRATEGY — Time-series variation across subperiods with differing realized default experience (e.g., 2008–2009, 2020) as natural stress tests.

EXPECTED CONTRIBUTION — Distinguishes a genuine near-zero default premium from a peso-problem artifact of the original sample.

MAIN RISK — Structural changes in corporate bond market liquidity and rating methodology over sixty years complicate direct comparison.

4. Momentum-augmented replication to isolate what the three-factor model still misses

RESEARCH QUESTION — On the original 1963–1991 sample and test assets, does adding a momentum factor (Carhart, 1997) further reduce the two remaining large intercepts identified in Section 6.1, or are they orthogonal to momentum?

MOTIVATION — The paper's own evidence leaves two of twenty-five stock intercepts economically large; whether momentum explains specifically those two is untested here.

HYPOTHESIS — The residual large intercepts are concentrated in portfolios with extreme recent-return histories, consistent with an omitted momentum factor rather than a size/value misspecification.

ECONOMIC MECHANISM — Momentum and value are known to be weakly negatively correlated characteristics; a portfolio extreme on one can be poorly spanned by a model built only for the other.

LITERATURE CONNECTION — Direct four-factor extension per Carhart (1997) and Jegadeesh and Titman (1993).

DATA — Same 25 size-BE/ME portfolios and sample window as the original paper, plus a constructed UMD momentum factor.

VARIABLES — Four-factor loadings and intercepts for the same 25 portfolios.

METHOD — Re-run equation (5) with UMD added; compare intercept magnitudes and counts directly against Table 9a.

IDENTIFICATION STRATEGY — Same time-series regression design as the original paper, changing only the factor set, for maximal comparability.

EXPECTED CONTRIBUTION — A precise, comparable answer to "what does the three-factor model still miss, on its own original test assets?"

MAIN RISK — Momentum factor construction conventions postdate 1991 and require care to avoid look-ahead bias when applied to the original sample window.

5. Characteristics versus covariances using modern panel methods

RESEARCH QUESTION — Using the full post-1993 sample, does size/BE/ME as a raw firm characteristic retain explanatory power for returns after controlling flexibly (not just linearly) for SMB/HML factor loadings, replicating the Daniel-Titman (1997) design with modern machine-learning-based characteristic controls?

MOTIVATION — This is the sharpest direct challenge to the paper's risk-factor interpretation (Section 10.2) and has not been revisited with modern nonparametric or ML tools.

HYPOTHESIS — Characteristics retain some independent explanatory power beyond covariances, but the share attributable to covariances has grown as the value and size factors became widely known and traded against.

ECONOMIC MECHANISM — If mispricing is partly self-correcting once discovered and exploited, the characteristic-versus-covariance split should shift over time in a way pure risk-based pricing would not predict.

LITERATURE CONNECTION — Direct extension of Daniel and Titman (1997), engaging the same identifying question this paper's Section 8.3 leaves open.

DATA — CRSP/Compustat 1963–2025, characteristic-sorted and covariance-sorted portfolios.

VARIABLES — Raw characteristics (size, BE/ME), rolling factor loadings, subsequent returns.

METHOD — Double-sort and Fama-MacBeth regressions controlling flexibly for factor loadings via machine-learning residualization, split pre/post factor "discovery" (approximately 1993 and 2013 publication milestones).

IDENTIFICATION STRATEGY — Structural break test around the dates the factors became widely known and investable via commercial factor funds.

EXPECTED CONTRIBUTION — Direct, time-varying evidence on the risk-versus-mispricing question central to this paper's interpretation, addressing the Section 11.2 literature-level gap.

MAIN RISK — Structural-break identification is confounded by the simultaneous growth of factor-investing assets under management and changes in market microstructure over the same period.

12.1 Ranking of the strongest ideas

EXTENSION Idea 4 (momentum-augmented replication on the original test assets) ranks highest on feasibility and methodological cleanliness: it reuses the paper's own test assets and design with a single, well-established additional factor, requires no novel data construction beyond a standard momentum portfolio, and yields a directly comparable answer. Idea 5 (characteristics versus covariances, modern panel methods) ranks highest on novelty and potential contribution, because it engages the paper's single most important unresolved question (Section 8.3) with tools unavailable in 1993 or even 1997, though it is more data- and computation-intensive and its structural-break identification is less clean. Idea 1 (full-universe breakpoints) ranks third: highly feasible and directly policy-relevant for anyone using SMB/HML today, but lower in novelty since breakpoint sensitivity has been informally discussed elsewhere in the literature. The **top three**, in order, are Idea 4, Idea 5, and Idea 1; Ideas 2 and 3, while economically well-motivated, depend on distress and default data that are noisier and less standardized, raising their execution risk relative to the top three.

13. Learning Framework

13.1 Prerequisite concepts

The Sharpe-Lintner CAPM and its single-beta prediction; the Merton (1973) Intertemporal CAPM and Ross (1976) Arbitrage Pricing Theory at the level of their reduced-form implications (not full derivations); ordinary least squares time-series regression, including interpretation of intercepts, slopes, R^2 , and heteroskedasticity/autocorrelation-robust standard errors; the mechanics of value-weighted portfolio construction and independent double-sorting.

13.2 Core concepts learned

The distinction between a characteristic that predicts average returns cross-sectionally and a factor-mimicking portfolio that captures common time-series variation, and why only the latter satisfies the definition of a risk factor under Merton/Ross; the Black-Jensen-Scholes time-series regression methodology as an alternative to Fama-MacBeth cross-sectional regression, including its comparative advantages (natural extension to multiple asset classes, a direct zero-intercept test) and its comparative weaknesses (vulnerability to in-sample factor construction); the specific mechanics of the SMB/HML sequential double-sort; the diagnostic value of decomposing a factor's explanatory power into "captures common variation" versus "carries a nonzero price of risk," illustrated by the DEF dissociation result.

13.3 Important equations

Equation (1'), the time-series regression test with its Merton (1973) zero-intercept implication; equations (2)–(3), the SMB and HML portfolio-construction formulas; equation (6), the orthogonalized-market-factor regression and its role in resolving the apparent stock-bond segmentation puzzle.

13.4 Methods learned

Sequential independent double-sorting to construct characteristic-mimicking zero-investment portfolios; time-series asset-pricing regression as a joint test of common variation and cross-sectional fit; orthogonalization of a collinear regressor as a diagnostic tool for disentangling overlapping factor exposures.

13.5 Questions to test understanding

COMPREHENSION

Can you explain, without notes, why the paper switches from the Fama-MacBeth cross-sectional regression approach of Fama and French (1992a) to the Black-Jensen-Scholes time-series approach, and what specific limitation of the former this switch is designed to overcome?

ANALYSIS

Can you explain why a factor (DEF) can produce loadings that are strongly, monotonically related to size and credit quality, more than seven standard errors from zero, and still fail to explain any of the cross-sectional pattern in average returns? What does this imply about the difference between "capturing common variation" and "being priced"?

RESEARCH

Design a single additional test, using only data and methods available to the original authors in 1993, that would have let them distinguish the risk-based interpretation of HML from the in-sample factor-mining alternative raised in Section 8.3. What is the strongest objection to your proposed test?

13.6 Papers to read next

Fama and French (1992a), "The Cross-Section of Expected Stock Returns" (the direct predecessor); MacKinlay (1995), "Multifactor Models Do Not Explain Deviations from the CAPM" (the sharpest methodological challenge); Fama and French (2015), "A Five-Factor Asset Pricing Model" (the direct lineal descendant).

14. Conclusion

One-sentence thesis — Fama and French (1993) show that portfolios built to mimic the empirically observed size and book-to-market effects in stock returns, together with term-structure and default-risk mimicking portfolios for bonds, behave like genuine common risk factors under the Black-Jensen-Scholes time-series test: they capture the great majority of shared variation in returns and leave average-return intercepts close to zero, providing the empirical foundation for reading these characteristics as priced risk rather than unexplained anomaly.

Three most important insights — first, a characteristic's cross-sectional predictive power for average returns and a mimicking portfolio's ability to capture common time-series variation are logically distinct properties, and the paper's central contribution is a design that tests both simultaneously rather than assuming one implies the other. Second, the market factor alone explains why stocks earn more than bills, while SMB and HML explain the dispersion of average returns across stocks — a clean, economically interpretable division of labor among the three factors. Third, a factor can have strong, well-determined loadings and still carry an economically trivial price of risk (the DEF dissociation), a result that is as instructive about the limits of factor-model testing as the paper's headline findings are about its successes.

Strongest evidence — the collapse of the twenty-five stock-portfolio intercepts toward zero when SMB and HML are added to the market regression (only two of twenty-five economically large, sixteen within 0.1% of zero), set against R^2 improvements from a market-alone range as low as 0.61–0.70 to a three-factor range of 0.94–0.97 for the smallest-size portfolios.

Biggest weakness — SMB and HML are constructed from, and tested on, the same sample and characteristics already known (from the authors' own prior work) to predict average returns, so the paper's joint test cannot, by itself, distinguish a genuine priced-risk interpretation from a well-fitted in-sample description of an already-known regularity.

Most important research gap — no economic identification of the state variable(s) SMB and HML are meant to proxy for, leaving the risk-versus-mispricing question that has driven three subsequent decades of asset-pricing research unresolved within the paper itself.

Best research opportunity — a momentum-augmented replication on the paper's own original twenty-five test assets (Research Idea 4, Section 12), which would give a directly comparable, low-execution-risk answer to what the three-factor model still misses on its own terms.

Three papers to read next — Fama and French (1992a); MacKinlay (1995); Fama and French (2015).

15. References

- Banz, R.W., 1981. The relationship between return and market value of common stocks. *Journal of Financial Economics* 9, 3–18.
- Banz, R.W., Breen, W.J., 1986. Sample-dependent results using accounting and market data: some evidence. *Journal of Finance* 41, 779–793.
- Basu, S., 1983. The relationship between earnings' yield, market value, and return for NYSE common stocks: further evidence. *Journal of Financial Economics* 12, 129–156.
- Bhandari, L.C., 1988. Debt/equity ratio and expected common stock returns: empirical evidence. *Journal of Finance* 43, 507–528.
- Black, F., Jensen, M.C., Scholes, M., 1972. The capital asset pricing model: some empirical tests. In: Jensen, M.C. (Ed.), *Studies in the Theory of Capital Markets*. Praeger, New York.
- Breeden, D.T., 1979. An intertemporal asset pricing model with stochastic consumption and investment opportunities. *Journal of Financial Economics* 7, 265–296.
- Carhart, M.M., 1997. On persistence in mutual fund performance. *Journal of Finance* 52, 57–82.
- Chan, K.C., Chen, N., Hsieh, D.A., 1985. An exploratory investigation of the firm size effect. *Journal of Financial Economics* 14, 451–471.
- Chen, N., Roll, R., Ross, S.A., 1986. Economic forces and the stock market. *Journal of Business* 59, 383–403.
- Daniel, K., Titman, S., 1997. Evidence on the characteristics of cross sectional variation in stock returns. *Journal of Finance* 52, 1–33.
- Fama, E.F., French, K.R., 1989. Business conditions and expected returns on stocks and bonds. *Journal of Financial Economics* 25, 23–49.
- Fama, E.F., French, K.R., 1992a. The cross-section of expected stock returns. *Journal of Finance* 47, 427–465.

- Fama, E.F., French, K.R., 1992b. The economic fundamentals of size and book-to-market equity. Working paper, University of Chicago and Yale University.
- Fama, E.F., French, K.R., 1993. Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics* 33, 3–56.
- Fama, E.F., French, K.R., 1996. Multifactor explanations of asset pricing anomalies. *Journal of Finance* 51, 55–84.
- Fama, E.F., French, K.R., 1998. Value versus growth: the international evidence. *Journal of Finance* 53, 1975–1999.
- Fama, E.F., French, K.R., 2015. A five-factor asset pricing model. *Journal of Financial Economics* 116, 1–22.
- Hou, K., Xue, C., Zhang, L., 2015. Digesting anomalies: an investment approach. *Review of Financial Studies* 28, 650–705.
- Jegadeesh, N., Titman, S., 1993. Returns to buying winners and selling losers: implications for stock market efficiency. *Journal of Finance* 48, 65–91.
- Keim, D.B., 1988. Stock market regularities: a synthesis of the evidence and explanations. In: Dimson, E. (Ed.), *Stock Market Anomalies*. Cambridge University Press.
- Kothari, S.P., Shanken, J., Sloan, R.G., 1995. Another look at the cross-section of expected stock returns. *Journal of Finance* 50, 185–224.
- Lakonishok, J., Shleifer, A., Vishny, R.W., 1994. Contrarian investment, extrapolation, and risk. *Journal of Finance* 49, 1541–1578.
- Lintner, J., 1965. The valuation of risk assets and the selection of risky investments in stock portfolios and capital budgets. *Review of Economics and Statistics* 47, 13–37.
- MacKinlay, A.C., 1995. Multifactor models do not explain deviations from the CAPM. *Journal of Financial Economics* 38, 3–28.
- Merton, R.C., 1973. An intertemporal capital asset pricing model. *Econometrica* 41, 867–887.
- Merton, R.C., 1980. On estimating the expected return on the market: an exploratory investigation. *Journal of Financial Economics* 8, 323–361.
- Reinganum, M.R., 1981. Misspecification of capital asset pricing: empirical anomalies based on earnings' yields and market values. *Journal of Financial Economics* 9, 19–46.
- Rosenberg, B., Reid, K., Lanstein, R., 1985. Persuasive evidence of market inefficiency. *Journal of Portfolio Management* 11, 9–16.
- Ross, S.A., 1976. The arbitrage theory of capital asset pricing. *Journal of Economic Theory* 13, 341–360.
- Shanken, J., 1982. The arbitrage pricing theory: is it testable? *Journal of Finance* 37, 1129–1140.
- Shanken, J., Weinstein, M.I., 1990. Macroeconomic variables and asset pricing: estimation and tests. Working paper, University of Rochester.
- Sharpe, W.F., 1964. Capital asset prices: a theory of market equilibrium under conditions of risk. *Journal of Finance* 19, 425–442.